

Inductive Learning

- It is called as ***Deterministic Supervised Learning***.
- It is also known as ***Discovery Learning*** as the observer discovers the rules (learning) by observing.
- The i/p 'x' given to function 'f' and the o/p is 'f(x)'
- **tries to find a function h (the *hypothesis*) that approximates a set of samples defining a function 'f'**
 - the samples are usually provided as input-output pairs $(x, f(x))$
- **relies on inductive inference, or induction**
 - conclusions are drawn from specific instances and rules are generated based on learning.

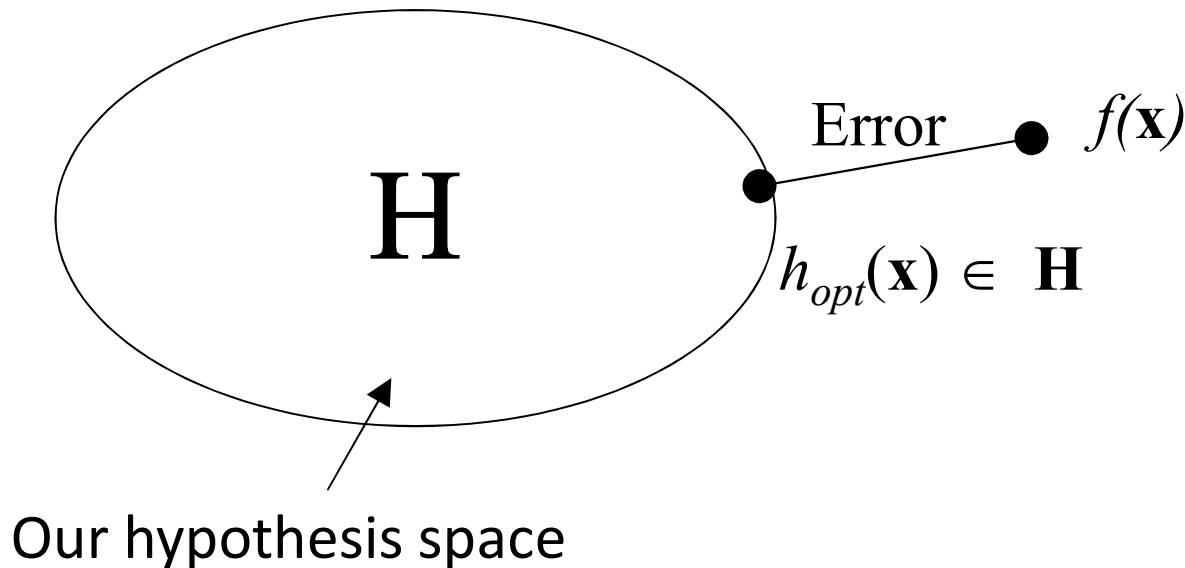
Hypotheses

- **Finding a suitable hypothesis can be difficult**
 - since the function f is unknown, it is hard to tell if the hypothesis h is a good approximation
- **The *hypothesis space* describes the set of hypotheses under consideration**
 - e.g. polynomials, sinusoidal functions, propositional logic, predicate logic, ...
 - the choice of the hypothesis space can strongly ***influence the task*** of finding a suitable function
 - while a very general hypothesis space (e.g. Turing machines) may be guaranteed to ***contain a suitable function***, it can be difficult to find it

Ockham's razor : if multiple hypotheses are consistent with the data, choose the simplest one

The idealized inductive learning problem

Find appropriate hypothesis space $\mathbf{H}$ and find $h(\mathbf{x}) \in \mathbf{H}$ with minimum “distance” to $f(\mathbf{x})$ (“error”)



The learning problem is realizable if $f(\mathbf{x}) \in \mathbf{H}$.

Inductive learning - example

- $f(\mathbf{x})$ is the **target function**
- An **example** is a pair $[\mathbf{x}, f(\mathbf{x})]$
- Learning task: find a **hypothesis** h such that $h(\mathbf{x}) \approx f(\mathbf{x})$ given a training set of examples $D = \{[\mathbf{x}_i, f(\mathbf{x}_i)]\}, i = 1, 2, \dots, N$
- **Example....**
 - Mango $\rightarrow f(\text{Mango}) \rightarrow \text{sweet} (e1)$*
 - Banana $\rightarrow f(\text{Banana}) \rightarrow \text{sweet}(e2)$*
 - ...*
 - Fruits $\rightarrow f(\text{Fruits}) \dots\dots\dots (general rule)$*